

tf.range

 View source on GitHub (https://github.com/tensorflow/tensorflow/blob/v2.16.1/tensorflow/python/ops/math_ops.py#L1940-L2021)

Creates a sequence of numbers.

```
tf.range(limit, delta=1, dtype=None, name='range')
tf.range(start, limit, delta=1, dtype=None, name='range')
```

Used in the notebooks

Used in the guide	Used in the tutorials
• DTensor concepts (https://www.tensorflow.org/guide/dtensor_overview) • Better performance with tf.function (https://www.tensorflow.org/guide/function) • Training checkpoints (https://www.tensorflow.org/guide/checkpoint) • Estimators (https://www.tensorflow.org/guide/estimator) • Optimizers with Core APIs (https://www.tensorflow.org/guide/core/optimizers_core)	• Integrated gradients (https://www.tensorflow.org/tutorials/interpretability/integrated_gradients) • DeepDream (https://www.tensorflow.org/tutorials/generative/deepdream) • Save and load a model using a distribution strategy (https://www.tensorflow.org/tutorials/distribute/save_and_load) • Playing CartPole with the Actor-Critic method (https://www.tensorflow.org/tutorials/reinforcement_learning/actor_critic) • Distributed Input (https://www.tensorflow.org/tutorials/distribute/input)

Creates a sequence of numbers that begins at `start` and extends by increments of `delta` up to but not including `limit`.

The dtype of the resulting tensor is inferred from the inputs unless it is provided explicitly.

Like the Python builtin `range`, `start` defaults to 0, so that `range(n) = range(0, n)`.

For example:

```
>>> start = 3
>>> limit = 18
>>> delta = 3
>>> tf.range(start, limit, delta)
<tf.Tensor: shape=(5,), dtype=int32,
numpy=array([ 3,  6,  9, 12, 15], dtype=int32)>
```

```
>>> start = 3
>>> limit = 1
>>> delta = -0.5
>>> tf.range(start, limit, delta)
<tf.Tensor: shape=(4,), dtype=float32,
numpy=array([3. , 2.5, 2. , 1.5], dtype=float32)>
```

```
>>> limit = 5
>>> tf.range(limit)
<tf.Tensor: shape=(5,), dtype=int32,
numpy=array([0, 1, 2, 3, 4], dtype=int32)>
```

Args

start	A 0-D Tensor (scalar). Acts as first entry in the range if limit is not None; otherwise, acts as range limit and first entry defaults to 0.
limit	A 0-D Tensor (scalar). Upper limit of sequence, exclusive. If None, defaults to the value of start while the first entry of the range defaults to 0.
delta	A 0-D Tensor (scalar). Number that increments start . Defaults to 1.
dtype	The type of the elements of the resulting tensor.
name	A name for the operation. Defaults to "range".

Returns
An 1-D Tensor of type dtype .

numpy compatibility

Equivalent to np.arange

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